

# SANJAY SURYA KANNATHASAN

AI/ML Associate Engineer · Software Engineering & Applied AI

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## PROFESSIONAL SUMMARY

Software engineer and AI/ML practitioner currently completing an **MSc in Data Science and Analytics at Maynooth University**. Combines strong **Python and Java backend engineering** with hands-on experience building **ML pipelines, REST APIs, and AI-powered applications**. Built a Claude-powered AI agent at the **Anthropic Hackathon 2025** and delivered production-quality ML models during an industry internship. Experienced with **LLM integration, prompt engineering, NLP, and data pipelines**. Eager to contribute to J.P. Morgan's enterprise AI/ML solutions at the intersection of rigorous software engineering and applied machine learning.

## EDUCATION

### MSc Data Science and Analytics

Sep 2025 – Sep 2026 | Maynooth, Ireland

**Maynooth University** — Statistical Machine Learning, Spatial Data Analytics, AI Applications, Data Visualisation

### B.E. Computer Science and Engineering (CGPA: 7.9 / 10)

2020 – 2024 | Chennai, India

**Panimalar Institute of Technology, Anna University**

## TECHNICAL SKILLS

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|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages      | Python (primary), Java, SQL, R, HTML/CSS, JavaScript (basic)                                                                                                          |
| AI / ML        | Scikit-learn, Pandas, NumPy, NLP, LLM APIs (Claude/OpenAI), Prompt Engineering, RAG Concepts, Embeddings, Feature Engineering, Predictive Modelling, Model Evaluation |
| Backend & APIs | Spring Boot, REST APIs, SMTP, MySQL, PostgreSQL, gRPC (conceptual), Postman                                                                                           |
| MLOps / DevOps | Git, CI/CD basics, MLflow (conceptual), Model Versioning, Testing (Unit/Integration), Agile/Scrum                                                                     |
| Cloud & Infra  | AWS (basic), Docker (working knowledge), Azure (exposure), Kubernetes (conceptual)                                                                                    |
| Tools          | Jupyter Notebook, VS Code, Eclipse, Anaconda, GitHub                                                                                                                  |

## KEY PROJECTS & AI EXPERIENCE

**CareCom — AI Agent · Anthropic Claude Hackathon** | Claude API · LLM Integration · Python · Prompt Engineering Mar 2025

🔗 **Anthropic Claude Hackathon 2025** · Group 17 · Track: Economic Empowerment & Education

- Designed and built CareCom — an LLM-powered AI agent using Anthropic's Claude API, implementing conversational AI workflows and prompt engineering to automate real-world economic tasks.
- Prototyped RAG-style information retrieval and agent-based task automation patterns within a competitive time constraint — directly aligned with J.P. Morgan's GenAI/Copilot workloads.
- Collaborated cross-functionally to define product scope, API integration strategy, and safety considerations for AI outputs; shipped a working demo in under 24 hours.

**AI Assistant Chatbot** | Python · NLP · Speech Recognition · REST APIs

2023

- Built an end-to-end NLP chatbot processing 500+ test queries with 92% contextual accuracy; integrated speech recognition, intent detection, and text-to-speech for human-like interaction.
- Designed modular Python architecture enabling rapid feature iteration; applied AI pipeline best practices (preprocessing, inference, output validation) to ensure reliability.
- Demonstrated applied LLM-adjacent engineering: intent classification, contextual response generation, and conversational state management — skills directly applicable to Copilot Agent development.

**Real Estate Price Forecasting · Production ML Pipeline** | Python · Scikit-learn · Regression · Pandas · Postman · REST API

2023

- Developed and evaluated regression-based ML models on 2,000+ records, achieving 88% prediction accuracy after full pipeline: data ingestion, cleaning, outlier detection, feature engineering, model selection, and evaluation.
- Deployed prediction functionality via a Postman-tested REST API; instrumented model outputs with accuracy metrics — demonstrating MLOps-adjacent lifecycle awareness (train → evaluate → deploy → monitor).
- Translated raw data into actionable business insights projecting 15% cost savings — showing ability to connect ML output to enterprise decision-making.

**EcoConstruct — MSc Dissertation (Data-Driven ML Visualiser)** | Python · ML Pipelines · Data Visualisation · Feature Engineering 2025–2026

- ▶ Building a sustainability intelligence platform for the construction sector; applies full ML lifecycle: dataset ingestion, feature engineering, model training (Scikit-learn), evaluation, and interactive visualisation dashboards.
- ▶ Supervised by Dr. Fergal Stapleton & Dr. Edgar Galván at Maynooth University; collaborates with academic stakeholders to translate domain requirements into production-quality data pipelines.

**Attendance System + Secure Email Authenticator** | Java · Spring Boot · MySQL · REST APIs · SMTP · OTP Security 2022–2023

- ▶ Engineered two production-quality backend systems: a full-stack RESTful Spring Boot attendance tracker (500+ users) and an SMTP OTP authenticator achieving 99% verification success across 1,000+ simulated transactions.
- ▶ Optimised MySQL queries reducing report generation from 10 min to under 1 min; reduced unauthorised access risk by 70%; wrote unit and integration tests to ensure reliability and maintainability — aligned with JPM enterprise engineering standards.

**INTERNSHIP EXPERIENCE**

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**Data Science Intern · Mindenious** Jul 2025 – Aug 2025 | Bengaluru, India

- ▶ Developed, tested and evaluated ML models for predictive analytics on 10,000+ record datasets using Python (Pandas, NumPy, Scikit-learn), improving model performance by 20% through iterative feature engineering and hyperparameter tuning.
- ▶ Built reusable Python scripts for data processing, experimentation workflows, and model training pipelines; created model validation metrics to assess predictive accuracy.
- ▶ Worked in an agile cross-functional team, presenting data-driven insights to business stakeholders — directly applicable to JPM's collaborative product delivery model.

**Associate Process Intern · Sutherland Global Services** Aug 2024 – Oct 2024 | Chennai, India

- ▶ Analysed client interaction data to identify workflow inefficiencies, contributing to a 25% reduction in response times while maintaining SLA compliance; strengthened analytical thinking in high-volume production environments.

**CERTIFICATIONS & DEVELOPMENT**

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**Backend Development — Java, Spring Boot, REST APIs** · FIIT Institute, Chennai

**Data Science & Machine Learning Projects** · Mindenious Industry Internship (Python, Scikit-learn, Pandas, Model Evaluation)

**AI & NLP Application Development** · Academic Project Work (Python, NLP, Speech Recognition, AI Pipelines)

**Anthropic Claude AI Hackathon — LLM Agent Development** · March 2025 · Economic Empowerment & Education Track

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Currently on Stamp 2 visa · Eligible for full-time employment from September 2026 (Stamp 1G) · Dublin, Ireland